Telecom complaint and service credit assistant

The system in one sentence

Help customers understand a telecom complaint and receive a verified service credit.

Key design calls

Code sets credit amounts. Rejected: Model arithmetic. Why: The same facts must yield the same credit.

Staff approve unclear or larger credits. Rejected: Open model authority. Why: Exceptions need a named owner.

Check uncertain payments before retry. Rejected: Blind second attempt. Why: Avoid paying the same incident twice.

Show progress in the customer portal. Rejected: Unowned background waits. Why: Customers see who owns the next step.

Who acts and who decides

Action	Autonomy
Read the selected active service	acts alone
Explain supported findings	suggests
Issue eligible service credits	acts alone
Approve larger or unclear credits	human decides
Handle cash refunds and invoice changes	human decides

Top risks by RPN

FM3 | Duplicate or concurrent credit | RPN 225

FM4 | Uncertain posting retried | RPN 216

FM7 | Supervisor or notice clock lost | RPN 200

FM2 | Stale policy or invoice | RPN 180

FM9 | Cancellation or denied approval bypassed | RPN 180

Launch blockers and owners

H1: Proposed pilot policy P1: verified... Owner: Billing policy owner.

H2: Account, service and billing adapters... Owner: Billing engineering owner.

H3: Proposed supervisor queue is staffed... Owner: Support operations owner.

H4: Proposed retention is 30 days... Owner: Privacy and security owner.

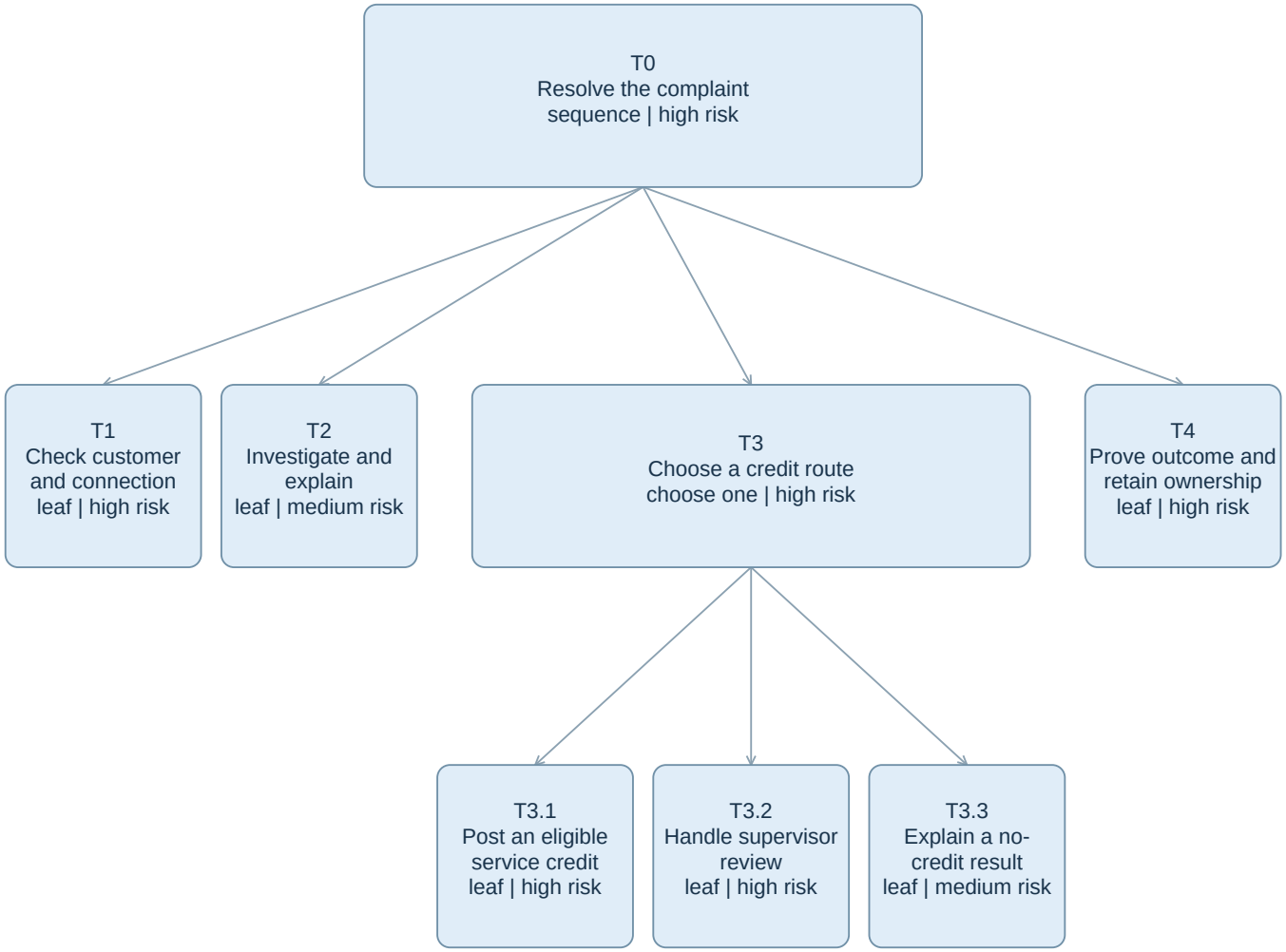
Illustrative cost per case

USD 0.00352 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: independent-agent; round 2; pass.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree



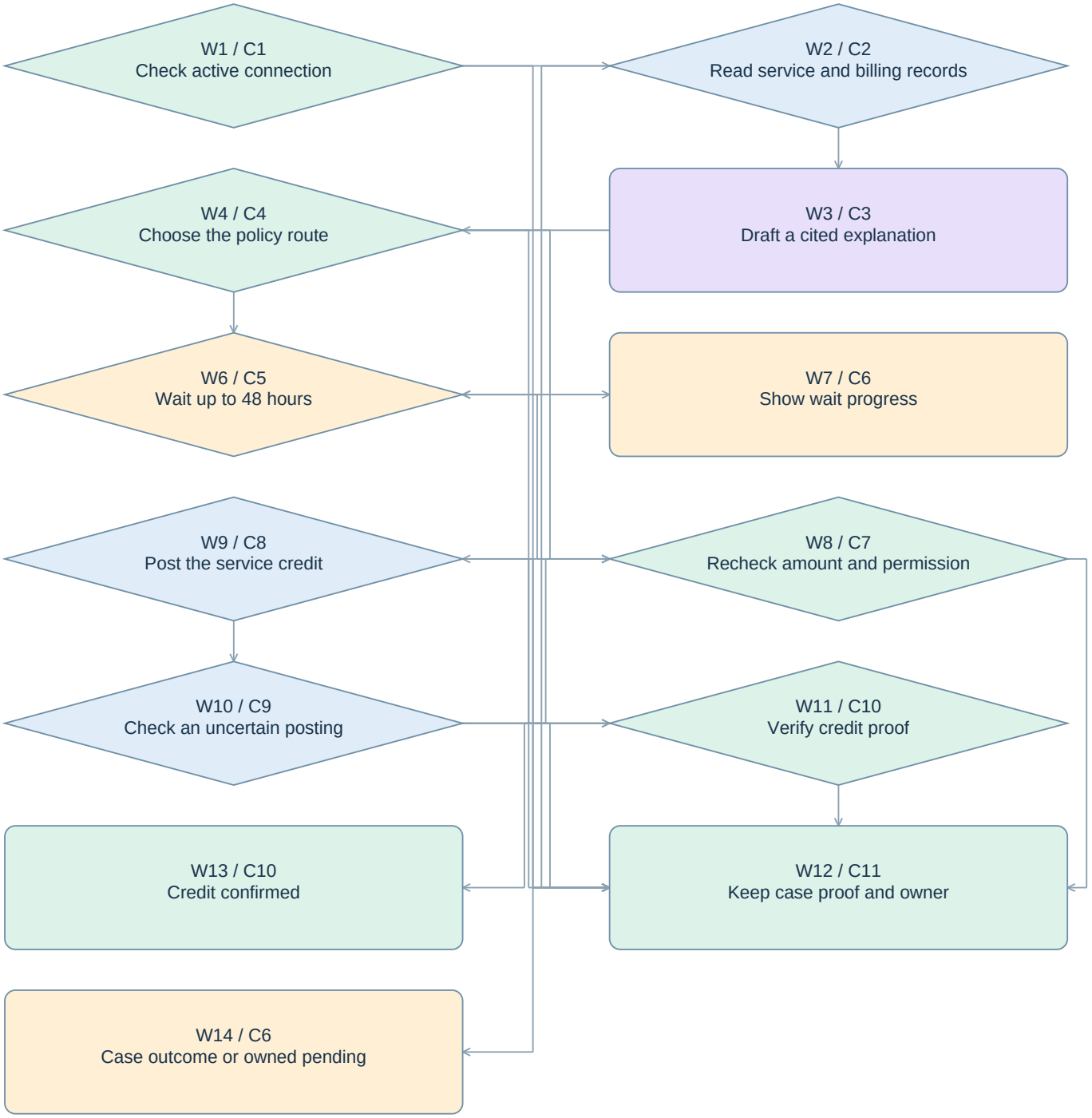
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

Start with verified outage credits. Investigate speed and billing complaints, then send unclear cases to staff. Keep pending cases owned.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	FM1	-	-	-	-	-	-	-
9	-	-	-	FM2, FM4, FM9	FM3	-	-	-	-	-
8	-	-	-	FM5, FM6	FM7	-	-	-	-	-
7	-	-	-	-	FM8	-	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Wrong or inactive connection	Authenticated scope first; explicit connection selection; atomic active/version check at commit.
FM2 Stale policy or invoice	Plan pins all versions, UTC interval, minor units, currency and financial allocation; API compare at commit.
FM3 Duplicate or concurrent credit	Unique business event key independent of complaint and amount; serialized entitlement; stable operationId; dedup queue events.
FM4 Uncertain posting retried	Never retry before definitive absent readback; same key once through C7. Ambiguity remains frozen/owned.
FM5 Model follows injected records	No model tools or financial outputs; isolate untrusted text, allowlist evidence citations, fallback template.
FM6 False credit completion	Receipt scope/amount checks and record guard before completed view; missing proof remains owned pending.
FM7 Supervisor or notice clock lost	Persist enqueueAt, notice sequence, queue owner and UTC deadlines; ENTRY/H24/H48 views; 1h duty-manager escalation.
FM8 Contradictory or missing evidence	No guessed entitlement. Provisional case; supervisor resolves authoritative records before exact plan approval. Unsupported external refund stays manual.
FM9 Cancellation or denied approval bypassed	Signed full immutable plan, event/case versions, expiry and cancellation precedence; unknown operation freezes later intent.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Authenticate and select	T1
C2 / Code: tools	Read complaint evidence	T2
C3 / Agents (LLM)	Explain supported findings	T2
C4 / Code: validation	Calculate and route	T3.1, T3.2, T3.3
C5 / Code: human triggers	Wait for supervisor decision	T3.2
C6 / Code: human triggers	Publish portal progress	T3.2, T4
C7 / Code: validation	Recheck posting authority	T3.1
C8 / Code: tools	Post one service credit	T3.1
C9 / Code: tools	Reconcile uncertain posting	T3.1, T4
C10 / Code: validation	Verify outcome proof	T4
C11 / Code: validation	Verify existing case proof	T3.3, T4

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - A durable code workflow

Choice: Use one durable code state machine with a bounded explanation model.

Rejected: Free-running multi-agent planner; synchronous request handler.

Trade-off: Code owns route/amount/clock. Planner adds authority risk; request handler cannot own 48h waits. Compare database-backed workers and workflow engine on durable timers...

F2 - State and evidence ownership

Choice: Relational case store for scope/versions/queue/event keys; immutable approved plans and append-only audit. Billing ledger remains financial truth.

Rejected: Vector store as authority; model conversation as state.

Trade-off: Retrieval may help explanation later, but cannot decide entitlement. Inbox events unique by provider eventId; DB compare-and-swap by caseVersion. Outbox and timers...

F3 - Separate read and credit identities

Choice: Gateway checks tenant/customer and active selected connection before reads; credit adapter only posts a scoped service-credit command.

Rejected: One broad billing admin credential.

Trade-off: Small blast radius; readback service identity may outlive portal session. Every financial writer enforces fresh plan and event key; human approval cannot...

Access: Scoped reads, credit-only write, separate durable reconciliation identity; portal ownership enforced on every GET.

F4 - Data deployment candidates

Choice: Choose managed relational DB, encrypted object audit and a secrets service in approved region; provider decision remains H4 spike.

Rejected: Cross-region unreviewed replication; embeddings over raw billing records.

Trade-off: AWS: RDS/object storage/secrets candidates; GCP: Cloud SQL/object storage/secret manager candidates; Azure: managed PostgreSQL/blob/key vault candidates; open source: PostgreSQL/S3-compatible storage/Vault candidates. Compare retention...

Access: Scoped reads, credit-only write, separate durable reconciliation identity; portal ownership enforced on every GET.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Model cost

Illustrative model cost is USD 0.00352 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV11	EV12
FM1	-	-	X	X	-	-	-	-	-	-	-	-
FM2	X	-	-	-	X	X	X	-	-	-	-	-
FM3	-	-	-	-	-	-	-	X	X	X	-	-
FM4	-	-	-	-	-	-	-	-	X	X	X	-
FM5	X	-	-	-	-	-	-	-	-	-	-	X
FM6	X	-	-	-	-	-	-	-	X	-	-	-
FM7	-	-	-	-	-	-	-	-	-	-	-	-
FM8	-	X	-	-	-	-	-	-	-	-	-	-
FM9	-	X	-	-	-	X	-	-	-	-	-	-
T1	X	-	X	X	-	-	-	-	-	-	-	-
T3.1	X	X	X	X	X	X	X	X	X	X	X	X
T3.2	-	X	-	-	-	X	-	-	-	-	-	-
T4	X	X	-	-	-	-	-	X	X	X	X	-

Evaluation coverage matrix - continued

Risk / task	EV13	EV14	EV15	EV16	EV17	EV18	EV19	EV20	EV24	EV25	EV26
FM1	-	-	-	-	-	-	-	-	-	X	-
FM2	-	-	-	-	-	X	-	-	X	-	X
FM3	-	-	-	-	-	-	-	-	X	-	-
FM4	-	-	-	-	-	-	-	-	-	-	-
FM5	-	-	-	-	-	-	-	-	-	-	-
FM6	X	-	-	-	-	-	-	-	-	-	-
FM7	-	X	X	X	-	-	-	-	-	-	-
FM8	-	-	-	-	X	-	X	-	-	-	-
FM9	-	-	-	X	-	-	-	X	-	-	X
T1	-	-	-	-	-	-	-	-	-	X	-
T3.1	-	-	-	X	-	X	-	X	X	X	X
T3.2	-	X	X	X	X	-	-	X	-	-	X
T4	X	X	X	-	-	-	X	X	-	-	-

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Eligible credit positive
EV2	golden	Supervisor larger-credit positive
EV3	golden	Wrong customer blocks reads
EV4	golden	Inactive service rejected
EV5	golden	Changed policy invalidates plan
EV6	golden	Approval covers whole plan
EV7	golden	Revoked authority prevents credit
EV8	golden	Two cases cannot duplicate one incident
EV9	golden	Timeout committed and duplicate callback
EV10	golden	Definitive absence permits one guarded retry
EV11	golden	Unknown commit stays frozen
EV12	golden	Injection cannot control money
EV13	golden	Success response without record is not completion
EV14	golden	Wait notices and overdue ownership
EV15	golden	Portal outage retains customer notice
EV16	golden	Cancellation wins before post
EV17	golden	Conflicting billing or speed evidence abstains
EV18	golden	Remaining charge caps credit
EV19	golden	Evidence supports no service credit
EV20	golden	Denied larger credit
EV21	offline	Offline evidence and adapter suite
EV22	online	Online safety and queue monitoring
EV23	release	Release complete suite
EV24	golden	Two incidents cannot exceed one invoice cap
EV25	golden	Clarify among multiple active services
EV26	golden	Tampered approval amount is isolated

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Measure durable complaint resolution

Measure: Seven-day reopen rate = complaints reopened within7d / complaints receiving a decision, by decision week. Track median and p95 elapsed hours to proven credit or owned decision. Financial correctness =...

Signals: EV21 citation precision/route agreement; EV22 unsafe effects, queue and notice lag; model cost per case

R2 - Shadow without customer effects

Phase / owner: Shadow: proposed two weeks after H4 data approval; no customer explanation or credit adapter permission. / Support quality owner

Gate: Entry: synthetic EV23 passes and lawful scoped read approvals. Exit: two weeks representative labelled traces covering outage/speed/billing and active/multi-connection strata; EV21 thresholds with adjudicated disagreements; no scoped data breach.

R3 - Assist staff with reviewed plans

Phase / owner: Assist: proposed two weeks; staff reviews every explanation and manually handles all financial commands. / Support operations owner

Gate: Entry: Shadow exit plus staff queue training and H1/H3 approvals. Exit: independently labelled plans meet EV21; H2 sandbox proves commit checks, uniqueness and readback; EV22 alarms exercised. No pilot effect required for...

R4 - Bounded automatic-credit pilot

Phase / owner: Pilot: verified outages only, INR under approved H1 cap; other complaints continue supervisor review. Proposed minimum four weeks plus7d follow-up. / Billing...

Gate: Entry: Assist exit, H1-H4 signed and Billing owner enables scoped adapter. Exit: powered R6 observation complete; zero unsafe financial effects, EV21 quality holds in each stratum, no unowned overdue/unknown case. Product and...

R5 - Expand only with evidence

Phase / owner: Scale: expand outage traffic in steps agreed by owners; speed/billing credits remain supervisor-only until a new verified policy/design review. / Product owner...

Gate: Entry: Pilot exit and owner decision. Weekly audit and EV23 on each policy/adapter change. Roll back new automatic traffic on any wrong-subject, duplicate or unproven completion; existing uncertain operations retain keys and...

R6 - Run a controlled complaint experiment

Test: No sample size, effect or savings invented. Pre-register analysis, attrition and missing-data handling; report per-stratum uncertainty.

Decision gate: Randomize authenticated customer account, sticky assignment, stratified by complaint type, language and connection count. Same household account never crosses arms; staff tool...

R7 - Own every pending case

Phase / owner: All phases / Support and Billing duty managers

Gate: Support owns supervisor waits and1h acknowledgement/48h decision; duty manager owns overdue. Billing owns uncertain posting, hourly read-only recovery and ledger reconciliation. Security owns scope/revocation incidents; model owner owns unsupported explanations. Product can...

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: A design and future test specification, not built telecom software.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	3/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

IR1 major resolved: Resolved in current iteration06: C8 malformed success routes UNKNOWN_COMMIT through W10/C9; explicit5/30/120s unresolved results lead C11-owned PENDING_RECONCILIATION. Expected trace forbids C10/W13, second POST and...

IR2 minor OPEN: Provider families and common comparison criteria are explicit, but no measured or scored tradeoff assessment is supplied. H4 keeps vendor/residency selection unresolved safely; complete...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.